Exp. No: 03

# Study of Different Types of Network Cables

a) Understand different types of network cables:

1. Cable type:- UTP (Unshielded twisted pair)

Characteristics:
- No shielding around wires
- Low cost and lightweight
- Easy to install & maintain.

Advantages / Disadvantages:
- Cheaper in cost
- Easy to install as they have a smaller overall diameter.
- More prone to EMI and noise.

Application:
- 10 Base-T Ethernet, Fast Ethernet, etc.

2. Cable type:- STP (Shielded Twisted Pair)

Characteristics:
- Twisted pair wires covered with metal shielding.
- Better protection against EMI
- More durable than UTP.

Advantages / Disadvantages:
- Shielded and faster than UTP
- Expensive, greater installation effort.

Application:
- Gigabit Ethernet, 10G Ethernet (55 m)

3. Cable Type: Coaxial cable

Characteristics:
- a central copper conductor with insulat
and outer shield.
- Good resistance to interface
- Higher bandwidth than twisted pair

Advantages / Disadvantages:
- High bandwidth
- Immune to interface.
- Limited distance, cost and size is bulk

Application:
- Speed signal is 100m Ethernet connec

4. Cable type: Fiber optic cable.

Characteristics:
- Uses light signals instead of electric
signals.
- Very high bandwidth and speed.
- Immune to electromagnetic interfe

Advantages / Disadvantages:
- High speed, bandwidth, security.
- Long distance.
- Expensive, require skilled installers

Application:
- Maximum distance of single fibre
cable is 100 Km.

b) Make Your Own Ethernet cross-over cable / Straight cable :-

T568B

White / Orange stripe
Orange Solid
White / green stripe
blue solid
white / blue stripe
green solid
white / brown stripe
brown

T568A

white / green stripe
green solid
white / orange stripe
white / brown stripe
brown solid
orange solid
blue solid
white / blue stripe.

12/9/24 9/10